

Qualification Pack

Chip Design Associate (O-Level Chip Design)

QP Code: NIE/ELE/Q0101

Version: 1.0

NSQF Level: 4

National Institute of Electronics and Information Technology (NIELIT) || NIELIT Bhawan, Plot No. 3, PSP
Pocket, Sector-8, Dwarka
New Delhi-110077 || email:puneetnema@nielit.gov.in

Qualification Pack

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NIE/ELE/Q0101: Chip Design Associate (O-Level Chip Design)

Brief Job Description

The Chip Design Associate is responsible for supporting the design and development of VLSI circuits, focusing on the implementation and verification of CMOS-based logic designs. They assist in Verilog RTL coding, synthesis, and performing static timing analysis to ensure proper functionality and performance. The role involves working with FPGA architectures, optimizing circuit layouts, and conducting simulations to validate designs. Additionally, they collaborate in projects involving the integration of VLSI components and participate in troubleshooting and debugging to meet design specifications.

Personal Attributes

A Chip Design Associate should have strong analytical and problem-solving skills to tackle complex VLSI design challenges. Attention to detail is crucial for ensuring precision in circuit design, verification, and timing analysis. They must be adaptable and willing to learn, keeping up with rapidly evolving technologies in the semiconductor industry. Effective communication and teamwork skills are also essential, as the role often involves collaborating with cross-functional teams on design projects.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [NIE/ELE/N0101: VLSI Fundamentals](#)
2. [NIE/ELE/N0102: Verilog RTL coding for Synthesis](#)
3. [NIE/ELE/N0103: Static Timing Analysis of VLSI Circuits](#)
4. [NIE/ELE/N0104: FPGA Architecture and Programming](#)
5. [NIE/ELE/N0112: Design and Implementation of VLSI Circuits Using Verilog and FPGA Technologies](#)
6. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	Electronics
Sub-Sector	Integrated Electronics and Circuits
Occupation	VLSI Design
Country	India

Qualification Pack

NSQF Level	4
Credits	15
Aligned to NCO/ISCO/ISIC Code	4
Minimum Educational Qualification & Experience	Previous relevant Qualification of NSQF Level (NSQF Level 3.5 in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with 1.5 years of experience OR Previous relevant Qualification of NSQF Level (NSQF Level 3 in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with 3 Years of experience OR 12th grade Pass (equivalent in Science with Physics and Maths) with NA of experience OR Completed 2nd year of the 3-year diploma after 10 (in Electronics and Communication Engineering/ Electrical Engineering/CS/IT and allied branches) with NA of experience
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NA
Minimum Job Entry Age	17 Years
Last Reviewed On	NA
Next Review Date	30/05/2027
NSQF Approval Date	30/05/2024
Version	1.0
Reference code on NQR	QG-04-EH-02593-2024-V1-NIELIT
NQR Version	1

Qualification Pack

NIE/ELE/N0101: VLSI Fundamentals

Description

VLSI Fundamentals provides a comprehensive overview of essential concepts in VLSI design, starting with an introduction to VLSI design principles and CMOS transistor theory. Participants will gain insights into CMOS inverter characteristics, logic design, and the intricacies of transistor-level schematics, layouts, and on-chip wire modeling. The course further delves into gate delay modeling, logical effort analysis, critical path optimization techniques, and timing analysis for sequential circuits. By the end of this course, participants will have a solid understanding of key fundamentals in VLSI design necessary for developing efficient and high-performance integrated circuits.

Scope

The scope covers the following :

- The scope of the course on VLSI Fundamentals is to provide a comprehensive understanding of the essential principles and practices in VLSI design. It covers various aspects crucial for designing integrated circuits, starting with an introduction to VLSI design principles and an in-depth exploration of CMOS transistor theory. Participants will learn about CMOS inverter characteristics and logic design, gaining insights into how to design basic logic gates and complex logic functions using CMOS technology.

Elements and Performance Criteria

Introduction to VLSI Design & CMOS Transistor Theory

To be competent, the user/individual on the job must be able to:

- PC1.** Describe the evolution and significance of VLSI technology, highlighting its impact on modern electronics and computing
- PC2.** Outline and differentiate between front-end and back-end VLSI design processes, and define MOSFET, explaining its operation principles and modes (cut-off, triode, saturation) along with their significance.

CMOS Inverter Characteristics and Logic Design

To be competent, the user/individual on the job must be able to:

- PC3.** Describe the functionality and design of CMOS implementations for basic logic gates (AND, OR, NAND, NOR), and design and analyze combinational logic circuits using CMOS gates, focusing on performance metrics.
- PC4.** Explain factors influencing inverter sizing and apply optimization techniques for performance and power consumption improvements, and combine basic CMOS gates to implement and optimize complex logic functions for speed or area efficiency.

Transistor Schematic, Layouts & On-chip wire modeling

To be competent, the user/individual on the job must be able to:

- PC5.** Develop transistor-level schematics for basic circuits and apply layout techniques to ensure manufacturability and performance, evaluating the impact of layout on circuit performance.
- PC6.** Explain design rules for different process nodes, optimize metal layer usage and interconnect routing for performance, and apply design rules and guidelines for efficient layout.

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Gate Delay, Logical Effort, Critical Path Optimization and Timing analysis for Sequential circuits

To be competent, the user/individual on the job must be able to:

- PC7.** Calculate gate delays based on MOSFET parameters, predict gate delays at different process nodes and operating conditions, and optimize logic gate designs for a balance between delay and power consumption using P/N ratio analysis.
- PC8.** Define logical effort and its relevance in circuit design, apply logical effort principles to estimate delay and power consumption, identify critical paths in combinational logic circuits, and apply gate sizing and logic restructuring techniques to optimize critical paths.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding the significance of VLSI technology and its role in modern electronics.
- KU2.** Analyzing noise margin and power consumption considerations in CMOS circuits.
- KU3.** Understanding schematic design, layout considerations, and design rules for integrated circuits.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Participants will hone their ability to analyze complex problems and devise solutions in VLSI design by understanding the significance of VLSI technology and the design flow. They will learn to apply critical thinking skills to evaluate noise margin, power consumption, and optimization techniques in CMOS inverter characteristics, logic design, and critical path optimization.
- GS2.** Learners will gain proficiency in using design tools and techniques essential for VLSI design, such as schematic design, layout considerations, and simulation of on-chip wires. They will develop skills in transistor-level design techniques, gate delay modeling, and logical effort analysis, enabling them to effectively navigate the complexities of CMOS transistor theory and logic design.
- GS3.** Participants will enhance their communication and collaboration skills. They will learn to effectively communicate their ideas, findings, and design solutions to peers and stakeholders. Collaboration on tasks such as bonding diagram design and packaging considerations will foster teamwork and the ability to work effectively in multidisciplinary settings within the semiconductor industry.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to VLSI Design & CMOS Transistor Theory</i>	25	20	-	5
PC1. Describe the evolution and significance of VLSI technology, highlighting its impact on modern electronics and computing	-	-	-	-
PC2. Outline and differentiate between front-end and back-end VLSI design processes, and define MOSFET, explaining its operation principles and modes (cut-off, triode, saturation) along with their significance.	-	-	-	-
<i>CMOS Inverter Characteristics and Logic Design</i>	25	20	-	5
PC3. Describe the functionality and design of CMOS implementations for basic logic gates (AND, OR, NAND, NOR), and design and analyze combinational logic circuits using CMOS gates, focusing on performance metrics.	-	-	-	-
PC4. Explain factors influencing inverter sizing and apply optimization techniques for performance and power consumption improvements, and combine basic CMOS gates to implement and optimize complex logic functions for speed or area efficiency.	-	-	-	-
<i>Transistor Schematic, Layouts & On-chip wire modeling</i>	25	20	-	5
PC5. Develop transistor-level schematics for basic circuits and apply layout techniques to ensure manufacturability and performance, evaluating the impact of layout on circuit performance.	-	-	-	-
PC6. Explain design rules for different process nodes, optimize metal layer usage and interconnect routing for performance, and apply design rules and guidelines for efficient layout.	-	-	-	-
<i>Gate Delay, Logical Effort, Critical Path Optimization and Timing analysis for Sequential circuits</i>	25	20	-	5

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC7. Calculate gate delays based on MOSFET parameters, predict gate delays at different process nodes and operating conditions, and optimize logic gate designs for a balance between delay and power consumption using P/N ratio analysis.	-	-	-	-
PC8. Define logical effort and its relevance in circuit design, apply logical effort principles to estimate delay and power consumption, identify critical paths in combinational logic circuits, and apply gate sizing and logic restructuring techniques to optimize critical paths.	-	-	-	-
NOS Total	100	80	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0101
NOS Name	VLSI Fundamentals
Sector	Electronics
Sub-Sector	
Occupation	VLSI Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

NIE/ELE/N0102: Verilog RTL coding for Synthesis

Description

The VLSI Design syllabus provides an introduction to VLSI technology, its applications, and the design flow. It covers the fundamentals of Register Transfer Level (RTL) design, including the use of Hardware Description Languages (HDLs) like Verilog or VHDL. Students will learn digital logic design principles, focusing on combinational and sequential logic, Boolean algebra, logic gates, flip-flops, registers, and state machines. Practical skills are developed through RTL design using HDL, simulation, and debugging, with a focus on functional verification techniques. Additionally, the course addresses timing constraints and analysis, teaching optimization techniques and resolving setup and hold time violations.

Scope

The scope covers the following :

- Gain a comprehensive understanding of VLSI technology, its significance, and applications, providing a solid foundation for further learning and understand the complete VLSI design flow, including the basics of Register Transfer Level (RTL) design, its process, and methodology.

Elements and Performance Criteria

RTL Design Methodology

To be competent, the user/individual on the job must be able to:

PC1. Understand the basics of Register Transfer Level (RTL) design.

PC2. Demonstrate proficiency in the RTL design process and methodology

Digital Logic Design Principles

To be competent, the user/individual on the job must be able to:

PC3. Apply combinational and sequential logic design concepts effectively.

PC4. Utilize Boolean algebra and logic gates to design digital circuits.

RTL Design Using HDL

To be competent, the user/individual on the job must be able to:

PC5. Demonstrate proficiency in Verilog syntax and constructs.

PC6. Design combinational and sequential logic circuits using HDL.

RTL Simulation and Verification

To be competent, the user/individual on the job must be able to:

PC7. Develop and utilize test benches for comprehensive testing.

PC8. Perform simulation and debugging of RTL designs to ensure functionality and correctness.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

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- KU1.** Grasping the essence of VLSI technology and its versatile applications across industries. Acquiring knowledge about the complete VLSI design flow, from conceptualization to fabrication, and its significance.
- KU2.** Developing a solid understanding of digital logic principles, including combinational and sequential logic.
- KU3.** Acquiring skills in functional verification techniques, test bench development, and timing analysis essential for robust RTL designs.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Developing problem-solving skills and analytical thinking to tackle complex RTL design challenges effectively
- GS2.** Enhancing communication skills for effectively conveying design ideas and collaborating with team members on RTL design projects
- GS3.** Cultivating attention to detail to ensure accuracy in RTL design, simulation, and verification, while also adapting to changing design requirements and methodologies

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>RTL Design Methodology</i>	25	20	-	5
PC1. Understand the basics of Register Transfer Level (RTL) design.	-	-	-	-
PC2. Demonstrate proficiency in the RTL design process and methodology	-	-	-	-
<i>Digital Logic Design Principles</i>	25	20	-	5
PC3. Apply combinational and sequential logic design concepts effectively.	-	-	-	-
PC4. Utilize Boolean algebra and logic gates to design digital circuits.	-	-	-	-
<i>RTL Design Using HDL</i>	25	20	-	5
PC5. Demonstrate proficiency in Verilog syntax and constructs.	-	-	-	-
PC6. Design combinational and sequential logic circuits using HDL.	-	-	-	-
<i>RTL Simulation and Verification</i>	25	20	-	5
PC7. Develop and utilize test benches for comprehensive testing.	-	-	-	-
PC8. Perform simulation and debugging of RTL designs to ensure functionality and correctness.	-	-	-	-
NOS Total	100	80	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0102
NOS Name	Verilog RTL coding for Synthesis
Sector	Electronics
Sub-Sector	
Occupation	VLSI Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

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NIE/ELE/N0103: Static Timing Analysis of VLSI Circuits

Description

This course provides a comprehensive overview of Static Timing Analysis (STA) in VLSI design, covering timing analysis fundamentals, including races, hazards, and setup and hold timing in both combinational and sequential circuits. Practical examples of setup and hold time violations and solutions are explored, alongside timing constraints for synthesis and circuit synthesis techniques. Advanced topics include timing performance improvement strategies such as pipelining and retiming, clock skew management, and open and closed-loop timing methodologies, with a focus on both block and chip level timing analysis. Students gain hands-on experience in Static Timing Analysis (STA) using Electronic Design Automation (EDA) tools, including timing analysis post-synthesis and place-and-route, interpretation of timing reports, Engineering Change Order (ECO) flows, and timing closure to meet sign-off requirements.

Scope

The scope covers the following :

- Gain insight into timing analysis principles for combinational and sequential circuits, including races, hazards, and setup/hold timing, with practical examples illustrating setup/hold time violations and their resolutions.
- Acquire knowledge of timing constraints essential for synthesis and circuit synthesis techniques, establishing a strong foundation in timing analysis fundamentals.
- Explore advanced timing performance enhancement techniques such as pipeline, retiming, and clock skew management, addressing timing analysis at both block and chip levels.
- Gain hands-on experience in Static Timing Analysis (STA) using Electronic Design Automation (EDA) tools, covering post-synthesis, place-and-route timing analysis, timing report interpretation, ECO flows execution, and achieving timing closure, ensuring practical proficiency in applying STA methodologies in real-world VLSI design projects.

Elements and Performance Criteria

Overview of VLSI STA

To be competent, the user/individual on the job must be able to:

- PC1.** Gain proficiency in Static Timing Analysis (STA) for VLSI design, encompassing analysis of timing in combinational and sequential circuits, resolution of timing issues like races and setup/hold violations through practical examples, and the establishment of synthesis-compatible timing constraints during circuit synthesis.

Timing performance

To be competent, the user/individual on the job must be able to:

- PC2.** Apply advanced techniques in timing performance enhancement, such as pipelining, retiming, and mitigation of clock skew issues, while conducting comprehensive timing analysis at block and chip levels to optimize design integrity and operational frequency across open and closed loop timing scenarios.

STA using EDA tools

To be competent, the user/individual on the job must be able to:

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- PC3.** Utilize Electronic Design Automation (EDA) tools to apply Static Timing Analysis (STA), ensuring thorough timing analysis post synthesis and place-and-route stages to validate design timing constraints. Evaluate timing reports, manage Engineering Change Orders (ECO) flows, and conduct sign-off checks to achieve timing closure.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Develop a comprehensive understanding of timing analysis principles, encompassing races, hazards, setup and hold timing, and maximum frequency of operation for both combinational and sequential circuits.
- KU2.** Recognize the importance of timing constraints in the synthesis process, gaining insight into circuit synthesis methodologies with a specific focus on timing analysis

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Ability to analyze complex timing issues in VLSI design, such as races, hazards, and setup/hold time violations, and devise effective solutions and Capacity to identify timing constraints and challenges during circuit synthesis and timing analysis, and apply analytical thinking to address them.
- GS2.** Proficiency in using Electronic Design Automation (EDA) tools for Static Timing Analysis (STA), including post-synthesis and place-and-route timing analysis, and interpreting timing reports.
- GS3.** Competence in executing Engineering Change Order (ECO) flows, conducting timing closure, and performing sign-off checks, demonstrating technical proficiency in utilizing EDA tools for VLSI design

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Overview of VLSI STA</i>	30	25	-	6
PC1. Gain proficiency in Static Timing Analysis (STA) for VLSI design, encompassing analysis of timing in combinational and sequential circuits, resolution of timing issues like races and setup/hold violations through practical examples, and the establishment of synthesis-compatible timing constraints during circuit synthesis.	-	-	-	-
<i>Timing performance</i>	30	25	-	7
PC2. Apply advanced techniques in timing performance enhancement, such as pipelining, retiming, and mitigation of clock skew issues, while conducting comprehensive timing analysis at block and chip levels to optimize design integrity and operational frequency across open and closed loop timing scenarios.	-	-	-	-
<i>STA using EDA tools</i>	40	30	-	7
PC3. Utilize Electronic Design Automation (EDA) tools to apply Static Timing Analysis (STA), ensuring thorough timing analysis post synthesis and place-and-route stages to validate design timing constraints. Evaluate timing reports, manage Engineering Change Orders (ECO) flows, and conduct sign-off checks to achieve timing closure.	-	-	-	-
NOS Total	100	80	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0103
NOS Name	Static Timing Analysis of VLSI Circuits
Sector	Electronics
Sub-Sector	
Occupation	VLSI Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

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NIE/ELE/N0104: FPGA Architecture and Programming

Description

This course offers a comprehensive study of Field-Programmable Gate Arrays (FPGAs), exploring their applications and advantages in digital design. Students delve into FPGA architecture, understanding its components like Look-Up Tables (LUTs) and flip-flops. They learn to write RTL code tailored for FPGA implementation through hands-on practice with Hardware Description Languages (HDLs) like Verilog. The curriculum covers the FPGA design flow from entry to bitstream generation, alongside design constraints. Advanced topics include FPGA design optimization techniques, verification, and debugging strategies, ensuring proficiency in FPGA-based projects.

Scope

The scope covers the following :

- The course will equip students with practical skills in Hardware Description Languages (HDLs), particularly Verilog, enabling them to proficiently write RTL code tailored for FPGA implementation. They will also gain a thorough understanding of the FPGA design flow, covering design entry, synthesis, placement, routing, and bitstream generation, alongside essential design constraints. Moreover, students will explore advanced optimization techniques like pipelining, retiming, and trade-offs between area, power, and performance to enhance FPGA designs. Additionally, they will learn FPGA verification and debugging methodologies, including simulation, testbench development, test vector creation, and effective debugging strategies, ensuring the development of robust and reliable FPGA designs.

Elements and Performance Criteria

Introduction to FPGAs

To be competent, the user/individual on the job must be able to:

PC1. Explain the fundamental concepts of FPGAs and their role in digital design.

PC2. Describe various applications of FPGAs in modern electronics

Hardware Description Languages for FPGAs

To be competent, the user/individual on the job must be able to:

PC3. Demonstrate knowledge of Verilog syntax and data types

PC4. Write effective RTL (Register Transfer Level) code for FPGA implementations.

FPGA Design Flow and Optimization Technique

To be competent, the user/individual on the job must be able to:

PC5. Implement design constraints effectively to meet specified requirements.

PC6. Apply optimization techniques such as timing analysis, pipelining, and retiming to improve FPGA performance

FPGA Verification and Debugging

To be competent, the user/individual on the job must be able to:

PC7. Apply debugging techniques to identify and resolve issues in FPGA designs

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- PC8.** Utilize simulation tools to ensure designs meet required specifications and performance criteria.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Develop a comprehensive understanding of FPGAs, including their applications and advantages in digital design, the internal architecture featuring components like Look-Up Tables (LUTs), flip-flops, and interconnects, as well as exploring different FPGA families and their available resources, such as memory blocks and DSP slices.
- KU2.** Master the use of Hardware Description Languages (HDLs), particularly Verilog, for writing RTL code for FPGAs, comprehend the complete FPGA design flow from design entry to bitstream generation, including the role of design constraints, and learn optimization techniques like pipelining, retiming, and balancing trade-offs between area, power, and performance to improve FPGA design efficiency
- KU3.** Grasp the importance of FPGA simulation and verification by learning to write effective test benches and test vectors, and acquire debugging skills using tools and techniques such as logic analyzers and on-chip debugging tools to ensure correct and optimized functionality of FPGA designs.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Gain comprehensive expertise in FPGA architecture, Verilog proficiency, and the complete FPGA design process, covering design entry to bitstream generation with the application of design constraints. Develop skills in analyzing and optimizing designs using techniques like pipelining and retiming, while balancing trade-offs between area, power, and performance.
- GS2.** Develop proficiency in FPGA design validation through simulation, leveraging effective test benches and test vectors. Master debugging tools and techniques to ensure the accuracy and optimization of FPGA functionality. Improve problem-solving skills and attention to detail by identifying and resolving design issues throughout the development process
- GS3.** Develop proficiency in managing FPGA projects from inception to implementation by applying systematic design approaches. Adapt to diverse FPGA families and resources, staying abreast of advancements in technology and tools. Enhance collaboration and communication skills through effective teamwork and clear articulation of design decisions and optimization strategies.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to FPGAs</i>	25	20	-	5
PC1. Explain the fundamental concepts of FPGAs and their role in digital design.	-	-	-	-
PC2. Describe various applications of FPGAs in modern electronics	-	-	-	-
<i>Hardware Description Languages for FPGAs</i>	25	20	-	5
PC3. Demonstrate knowledge of Verilog syntax and data types	-	-	-	-
PC4. Write effective RTL (Register Transfer Level) code for FPGA implementations.	-	-	-	-
<i>FPGA Design Flow and Optimization Technique</i>	25	20	-	5
PC5. Implement design constraints effectively to meet specified requirements.	-	-	-	-
PC6. Apply optimization techniques such as timing analysis, pipelining, and retiming to improve FPGA performance	-	-	-	-
<i>FPGA Verification and Debugging</i>	25	20	-	5
PC7. Apply debugging techniques to identify and resolve issues in FPGA designs	-	-	-	-
PC8. Utilize simulation tools to ensure designs meet required specifications and performance criteria.	-	-	-	-
NOS Total	100	80	-	20

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National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0104
NOS Name	FPGA Architecture and Programming
Sector	Electronics
Sub-Sector	
Occupation	VLSI Design
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

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NIE/ELE/N0112: Design and Implementation of VLSI Circuits Using Verilog and FPGA Technologies

Description

This NOS provides a comprehensive exploration of VLSI design principles, focusing on the use of Verilog for RTL coding and FPGA for circuit implementation. Students will learn the complete design flow, including simulation, synthesis, and optimization techniques while gaining practical experience through hands-on projects. Emphasizing key concepts such as CMOS technology, timing analysis, and verification, the course equips learners with the skills necessary to develop efficient and robust VLSI circuits.

Scope

The scope covers the following :

- The scope of this NOS encompasses a thorough understanding of VLSI design methodologies, including both theoretical foundations and practical applications. Participants will delve into CMOS transistor theory, logic design, and timing analysis while gaining hands-on experience in RTL coding with Verilog and circuit implementation on FPGAs. The course also covers crucial aspects of static timing analysis and design optimization, preparing students for real-world challenges in VLSI development.

Elements and Performance Criteria

Design and Implementation of VLSI Circuits Using Verilog and FPGA Technologies

To be competent, the user/individual on the job must be able to:

- PC1.** Participants will gain proficiency in writing RTL code using Verilog, learning the syntax, constructs, and design principles necessary for creating combinational and sequential logic circuits, as well as developing test benches for simulation and verification.
- PC2.** Demonstrate a clear understanding of the VLSI design flow by successfully outlining and explaining each stage, including front-end and back-end processes.
- PC3.** Develop and execute test benches to simulate RTL designs, successfully identifying and correcting errors or issues through debugging techniques.
- PC4.** Write clear, efficient, and syntactically correct RTL code for a variety of digital circuits, including both combinational and sequential logic components.
- PC5.** Integrate design principles such as modularity and reusability in Verilog code, demonstrating the ability to create scalable and maintainable digital systems.
- PC6.** Implement VLSI circuits on an FPGA, successfully configuring the design flow from synthesis to bitstream generation while adhering to design constraints.
- PC7.** Define and apply appropriate timing constraints during the FPGA design process, ensuring that the implemented circuit meets specified performance requirements.
- PC8.** Evaluate the performance of the implemented FPGA design, analyzing metrics such as power consumption, area, and timing to make informed optimization decisions.

Knowledge and Understanding (KU)

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The individual on the job needs to know and understand:

- KU1.** Understand the principles and significance of VLSI technology and its applications in modern electronic systems
- KU2.** Familiarity with writing and optimizing RTL code, as well as developing effective test benches for simulation and verification.
- KU3.** Knowledge of optimization techniques specific to FPGA design, including timing constraints, pipelining, and trade-offs between area, power, and performance
- KU4.** Comprehend timing constraints and their effects on the synthesis and performance of digital designs.

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** Apply critical thinking to optimize performance, power, and area in FPGA-based designs.
- GS2.** Work effectively in teams to handle collaborative VLSI design projects, fostering cooperation and knowledge-sharing.
- GS3.** Communicate effectively with peers and mentors, using appropriate technical terminology related to Verilog, FPGAs, and circuit design.
- GS4.** Adapt to new design tools and methodologies in VLSI and FPGA development environments.

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Design and Implementation of VLSI Circuits Using Verilog and FPGA Technologies</i>	-	-	80	20
PC1. Participants will gain proficiency in writing RTL code using Verilog, learning the syntax, constructs, and design principles necessary for creating combinational and sequential logic circuits, as well as developing test benches for simulation and verification.	-	-	-	-
PC2. Demonstrate a clear understanding of the VLSI design flow by successfully outlining and explaining each stage, including front-end and back-end processes.	-	-	-	-
PC3. Develop and execute test benches to simulate RTL designs, successfully identifying and correcting errors or issues through debugging techniques.	-	-	-	-
PC4. Write clear, efficient, and syntactically correct RTL code for a variety of digital circuits, including both combinational and sequential logic components.	-	-	-	-
PC5. Integrate design principles such as modularity and reusability in Verilog code, demonstrating the ability to create scalable and maintainable digital systems.	-	-	-	-
PC6. Implement VLSI circuits on an FPGA, successfully configuring the design flow from synthesis to bitstream generation while adhering to design constraints.	-	-	-	-
PC7. Define and apply appropriate timing constraints during the FPGA design process, ensuring that the implemented circuit meets specified performance requirements.	-	-	-	-
PC8. Evaluate the performance of the implemented FPGA design, analyzing metrics such as power consumption, area, and timing to make informed optimization decisions.	-	-	-	-
NOS Total	-	-	80	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	NIE/ELE/N0112
NOS Name	Design and Implementation of VLSI Circuits Using Verilog and FPGA Technologies
Sector	Electronics
Sub-Sector	
Occupation	VLSI Design
NSQF Level	4
Credits	5
Version	1.0
Last Reviewed Date	30/05/2024
Next Review Date	30/05/2027
NSQC Clearance Date	30/05/2024

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

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- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

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PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

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- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

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Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

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Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

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National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	18/02/2025
Next Review Date	18/02/2028
NSQC Clearance Date	18/02/2025

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

Assessment of the qualification evaluates candidates to ascertain that they can integrate knowledge, skills and values for carrying out relevant tasks as per the defined learning outcomes and assessment criteria. The underlying principle of assessment is fairness and transparency. The evidence of the outcomes and assessment criteria. Competence acquired by the candidate can be obtained by conducting Theory (Online), Practical assessment, internal assessment, Project/Presentation/ Assignment, Major Project. The emphasis is on the practical demonstration of skills & knowledge gained by the candidate through the training. Each OUTCOME is assessed & marked separately. A candidate is required to pass all OUTCOMES individually based on the passing criteria.

About Examination Pattern:

1. The question papers for the theory exams are set by the Examination wing (assessor) of NIELIT HQS.
2. The assessor assigns the roll number.
3. The assessor carries out theory online assessments through remote proctoring methodology. Theory examination would be conducted online and the paper comprise of MCQ. Conduct of assessment are through trained proctors. Once the test begins, remote proctors have full access to candidate's video feeds

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and computer screens. Proctors authenticate the candidate based on registration details, pre-test image captured and I- card in possession of the candidate. Proctors can chat with candidates or give warnings to candidates. Proctors can also take screenshots, terminate a specific user's test session, or re-authenticate candidates based on video feeds.

4. An External Examiner/ Observer may be deployed including NIELIT officials for evaluation of Practical examination/ internal assessment / Project/ Presentation/. Major Project (if applicable) would be evaluated preferably by external/ subject expert including NIELIT officials.

5. Pass percentage would be 50% marks in each component.

6. Candidates may apply for re-examination within the validity of registration (only in the assessment component in which the candidate failed).

7. For re-examination prescribed examination fee is required to be paid by the candidate only for the assessment component in which the candidate wants to reappear.

8. There would be no exemption for any paper/module for candidates having similar qualifications or skills.

9. The examination will be conducted in English language only.

Quality assurance activities: A pool of questions is created by a subject matter expert and moderated by other SME. Test rules are set beforehand. Random set of questions which are according to syllabus appears which may differ from candidate to candidate. Confidentiality and impartiality are maintained during all the examination and evaluation processes.

Minimum Aggregate Passing % at QP Level : 50

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0101.VLSI Fundamentals	100	80	0	20	200	21
NIE/ELE/N0102.Verilog RTL coding for Synthesis	100	80	0	20	200	21
NIE/ELE/N0103.Static Timing Analysis of VLSI Circuits	100	80	0	20	200	21

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National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
NIE/ELE/N0104.FPGA Architecture and Programming	100	80	0	20	200	21
NIE/ELE/N0112.Design and Implementation of VLSI Circuits Using Verilog and FPGA Technologies	0	0	80	20	100	11
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	5
Total	420	350	80	100	950	100

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Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
AA	Assessment Agency
AB	Awarding Body
ISCO	International Standard Classification of Occupations
NCO	National Classification of Occupations
NCrF	National Credit Framework
NOS	National Occupational Standard(s)
NQR	National Qualification Register
NSQF	National Skills Qualification Framework
OJT	On the Job Training

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Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

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Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
National Occupational Standard	NOS defines the measurable performance outcomes required from an individual engaged in a particular task. They list down what an individual performing that task should know and also do.
Qualification	A formal outcome of an assessment and validation process which is obtained when a competent body determines that an individual has achieved learning outcomes to given standards
Qualification File	A Qualification File is a template designed to capture necessary information of a Qualification from the perspective of NSQF compliance. The Qualification File will be normally submitted by the awarding body for the qualification.
Sector	A grouping of professional activities on the basis of their main economic function, product, service or technology.
Long Term Training	Long-term skilling means any vocational training program undertaken for a year and above. https://ncvet.gov.in/sites/default/files/NCVET.pdf